

Answer_Assignment_ AREA_MENSURATION

Exercise

1. **Ans: 576 cm²**; Area = $\frac{1}{2} \times b \times h = \frac{1}{2} \times 32 \times 36 = 576 \text{ cm}^2$

2. **Ans: 45 cm**;

Area = $\frac{1}{2} \times b \times h \Rightarrow 855 = \frac{1}{2} \times 38 \times h \Rightarrow h = 45 \text{ cm}$

3. **Ans: 900 cm²**; Area = $\frac{1}{2} \times b \times h = \frac{1}{2} \times 45 \times 40 = 900 \text{ cm}^2$

4. **Ans: 36 cm**; Area = $\frac{1}{2} \times b \times h \Rightarrow 864 = \frac{1}{2} \times b \times 48$

$\Rightarrow b = 36 \text{ cm}$

5. **Ans: 25 cm, 40cm**; Let $b = 5x$; $h = 8x$

Area = $\frac{1}{2} \times 5x \times 8x \Rightarrow (500 \times 2)/40 = x^2$

$x = 5$; $b = 5 \times 5 = 25 \text{ cm}$, $\Rightarrow h = 8 \times 5 = 40 \text{ cm}$

6. **Ans: 20 cm, 32cm**; Let $b = 5x$; $h = 8x$

Area = $\frac{1}{2} \times 5x \times 8x \Rightarrow (320 \times 2)/40 = x^2$

$\Rightarrow x = 4$; $b = 5 \times 4 = 20 \text{ cm}$, $\Rightarrow h = 8 \times 4 = 32 \text{ cm}$

7. a) **Ans: 216 cm²**; Area = $\frac{1}{2} \times b \times h = \frac{1}{2} \times 18 \times 24 = 216 \text{ cm}^2$

b) **Ans : 28.8 cm**; Area = $\frac{1}{2} \times b \times h \Rightarrow 216 = \frac{1}{2} \times 15 \times h$

$\Rightarrow h = 28.8 \text{ cm}$

8. **Ans: 126 cm²**;

Area = $\frac{1}{2} \times b \times h \Rightarrow \text{Area} = \frac{1}{2} \times 14 \times 18 = 126 \text{ cm}^2$

b) **Ans : 21 cm**; Area = $\frac{1}{2} \times b \times h \Rightarrow 126 = \frac{1}{2} \times 12 \times h$

$\Rightarrow h = 21 \text{ cm}$

9. **Ans: 28 cm and 35 cm**; Let $b = 4x$; $h = 5x$

Area = $\frac{1}{2} \times 4x \times 5x \Rightarrow (490 \times 2)/20 = x^2$

$x = 7 \Rightarrow B = 4 \times 7 = 28 \text{ cm}$, $\Rightarrow H = 5 \times 7 = 35 \text{ cm}$

10. **10 cm and 25 cm**; Let $b = 2x$; $h = 5x$

Area = $\frac{1}{2} \times 2x \times 5x \Rightarrow (125 \times 2)/10 = x^2$

$x = 5 \Rightarrow B = 2 \times 5 = 10 \text{ cm}$, $\Rightarrow H = 5 \times 5 = 25 \text{ cm}$

11. **20 cm² and 96 cm²**; $h^2 = 16^2 + 12^2 = 256 + 144 = 400$

$\Rightarrow H = 20 \text{ cm}^2 \therefore \text{Area} = \frac{1}{2} \times b \times h = \frac{1}{2} \times 16 \times 12 = 96 \text{ cm}^2$

12. **Ans: 15m, 17m**; Area = $\frac{1}{2} \times b \times h \Rightarrow 60 = \frac{1}{2} \times 8 \times h$

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$$\Rightarrow h \text{ (one leg)}=15\text{m} \quad \Rightarrow h^2 = 15^2 + 8^2 = 225+64= 289$$

$$\Rightarrow h \text{ (hypotenuse)}=17\text{m}$$

13. **Ans: 36:25;** Area 1 : Area 2 = $\frac{1}{2} \times b_1 \times h_1 : \frac{1}{2} \times b_2 \times h_2$

$$\Rightarrow 6 : 5 = 5 \times h_1 : 6 \times h_2 \Rightarrow 36 : 25 = h_1 : h_2$$

14. **Ans: 200000 m², Rs.600;**

$$\text{Area} = \frac{1}{2} \times b \times h = \frac{1}{2} \times 800 \times 500 = 200000$$

$$\therefore \text{Cost} = (30 \times 200000) / 10000 = \text{Rs.600}$$

15. **Ans: $9\sqrt{3} \text{ m}^2$;** Area of an equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

$$= \frac{\sqrt{3}}{4} \times 6 \times 6 = 9\sqrt{3} \text{ m}^2$$

16. **Ans: $16\sqrt{3} \text{ m}^2$;** Perimeter of an equilateral triangle = $3a$

$$\Rightarrow 24 \text{ m} = 3a \quad \Rightarrow a = 8 \text{ m.}$$

$$\text{Area of an equilateral triangle} = \frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} \times 8 \times 8 = 16\sqrt{3} \text{ m}^2$$

17. **Ans: 120 cm;** Perimeter = $x + x + 60$

$$\Rightarrow 300 - 60 = 2x; \quad \Rightarrow x = 120 \text{ cm}$$

18. **Ans: 150 cm²;** $l = \sqrt{h^2 - b^2} = \sqrt{25^2 + 20^2} = \sqrt{625 - 400} \quad \Rightarrow l = \sqrt{225} \quad \Rightarrow l = 15 \text{ m}$

$$\text{Since, length of the triangle is the height of the triangle, Area} = \frac{1}{2} \times b \times h = \frac{1}{2} \times 20 \times 15 = 150 \text{ cm}^2$$

19. **Ans: $16\sqrt{3} \text{ cm}^2$, 24 cm;**

$$\text{Area of an equilateral triangle} = \frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} \times 8 \times 8 = 16\sqrt{3} \text{ m}^2$$

$$\therefore \text{Perimeter of an equilateral triangle} = 3a = 3 \times 8 = 240 \text{ cm.}$$

20. **Ans: $60\sqrt{7} \text{ cm}^2$;**

$$\text{Let sides of the triangle be } 4x, 5x \text{ and } 6x$$

$$\therefore 4x + 5x + 6x = 60 \Rightarrow 15x = 60 \quad \therefore x = 4$$

$$\therefore \text{three sides of triangle are } 16\text{cm, } 20 \text{ cm and } 24 \text{ cm}$$

$$\therefore \text{Area} = \sqrt{s(s-a)(s-b)(s-c)} = \Rightarrow \sqrt{30(30-16)(30-20)(30-24)}$$

$$= 60\sqrt{7} \text{ cm}^2$$

21. **Ans: 300 m²;**

$$b = \sqrt{h^2 - l^2} = \sqrt{25^2 - 20^2} = \sqrt{625 - 400} \quad \Rightarrow b = \sqrt{225} \quad \Rightarrow b = 15 \text{ m}$$

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$$\text{Area} = b \times l = 20 \times 15 = 300 \text{ cm}^2$$

22. **Ans: 27 m²**; Let $b = x$ $\therefore l = 3x$ $\therefore \text{Perimeter} = 2(l + b)$

$$\Rightarrow 24 = 2(x + 3x) \quad \Rightarrow 12 = 4x; \quad \Rightarrow x = 3$$

$$\therefore b = 3\text{m and } l = 9\text{m} \therefore \text{Area} = l \times b = 3 \times 9 = 27 \text{ m}^2$$

23. **Ans: 2400 m²**; Perimeter = $2(l + b)$

$$\Rightarrow 200 = 2(l + 40) \quad \Rightarrow 100 - 40 = l \Rightarrow l = 60$$

$$\text{Area} = l \times b = 40 \times 60 = 2400 \text{ m}^2.$$

24. **Ans: 16 cm²**;

25. **Ans: 12 cm²**; Let $b = x$ $\therefore l = x + 1$

$$\therefore \text{Perimeter} = 2(x + x + 1) \Rightarrow 14 = 2(x + x + 1)$$

$$\Rightarrow 7 - 1 = 2x \quad \Rightarrow x = 3$$

$$\therefore b = 3\text{m and } l = 4\text{m} \therefore \text{Area} = l \times b = 3 \times 4 = 12 \text{ cm}^2$$

26. **Ans: $40/\sqrt{3}\text{m}$, $50/\sqrt{3}\text{m}$** ; Let the length = $5x$ & breadth = $4x$.

$$\begin{aligned} \text{Now, area} &= \frac{1}{15} \times 10,000 \text{ m}^2 = \frac{2000}{3} \text{ m}^2 \\ \text{so, } 5x \times 4x &= \frac{2000}{3} \Rightarrow 20x^2 = \frac{2000}{3} \Rightarrow x^2 = \frac{100}{3} \Rightarrow x = \frac{10}{\sqrt{3}} \\ \therefore \text{Length} &= 5x = \frac{50}{\sqrt{3}} \text{ m and Breadth} = 4x = \frac{40}{\sqrt{3}} \text{ m} \end{aligned}$$

27. **Ans: 2800 cm²**; Let $b = x \text{ cm}$ $\therefore l = 175x \text{ cm}$

$$\text{Now, Difference between } l \text{ \& } b = 175x - 100x = 75x$$

$$\Rightarrow 30 \text{ cm} = 75x \quad \Rightarrow x = \frac{30}{75} = x = \frac{2}{5} \text{ cm}$$

$$\therefore b = 100x = \frac{2}{5} \times 100 = 40 \text{ cm and } l = 175x = \frac{2}{5} \times 175 = 70 \text{ cm}$$

$$\therefore \text{Area} = l \times b \Rightarrow 70 \times 40 = 2800 \text{ cm}^2$$

28. **Ans: 40 cm**; Let $b = x$ and $l = 4x$

$$\therefore \text{Original area} = 4x \times x = 4x^2$$

$$\text{Now, New Area} = (4x + 10) \times (x + 10)$$

$$\therefore (4x + 10) \times (x + 10) - 4x^2 = 600$$

$$\Rightarrow 50x = 600 - 100 \quad \Rightarrow x = 10$$

$$\therefore l = 4 \times 10 = 40 \text{ cm}$$

29. **Ans. 70m**; Area of the room = $25 \times 14 = 350 \text{ m}^2$

$$\therefore \text{area of the carpet} = 350 \text{ m}^2$$

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$$\text{Length of Carpet} = \frac{\text{Area of Room}}{\text{Breadth of Carpet}} = \frac{350}{5} = 70\text{m}$$

30. Ans: 616 cm² ;

Largest circle which can be drawn will have a diameter of 28cm (as 28cm is the smallest side of the rectangle)

$$\therefore \text{Radius } \frac{28}{2} = 14\text{cm}$$

$$\therefore \text{Area of circle} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$$

31. Ans: 20 m; Let $b = 10x$ and $l = 14x$

$$\begin{aligned} \text{Area} &= 14x \times 10x = 560 & \Rightarrow 140x^2 &= 560 & \Rightarrow x^2 &= 4 \\ \Rightarrow x &= 2 & \therefore b &= 10 \times 2 = 20\text{m} \end{aligned}$$

32. Ans: 30 cm; Area = $l \times b \Rightarrow 360 = l \times b \Rightarrow l = \frac{360}{b}$

$$\text{Perimeter} = 2(l + b) \Rightarrow 7b = 2\left(\frac{360}{b} + b\right)$$

$$\Rightarrow 7b = \frac{720}{b} + 2b \quad \Rightarrow 5b^2 = 720 \quad \Rightarrow b^2 = 144$$

$$\Rightarrow b = 12 \quad \therefore l = \frac{360}{12} = 30\text{cm}$$

33. Ans: 30 cm; Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$

$$\Rightarrow 210 = \frac{1}{2} \times 14 \times d_2 \quad \Rightarrow d_2 = 30 \text{ cm}$$

34. Ans: 40 cm; Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$

$$\Rightarrow 320 = \frac{1}{2} \times 16 \times d_2 \quad \Rightarrow d_2 = 40 \text{ cm}$$

35. Ans: 96 cm², 40 cm;

$$\text{Area of rhombus} = \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 16 \times 12 = 96 \text{ cm}^2$$

$$\Rightarrow a^2 = (8)^2 + (6)^2 = 64 + 36 = 100 \quad \Rightarrow a = 10 \text{ cm}$$

$$\therefore \text{Perimeter of the rhombus} = 4a = 4 \times 10 = 40 \text{ cm.}$$

36. Ans: 616 m², 88 m;

$$\text{Area of a circle} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616 \text{ m}^2 \text{ and}$$

$$\text{Circumference of a circle} = 2\pi r = 2 \times \frac{22}{7} \times 14 = 88 \text{ m}$$

37. Ans: Rs. 6,528;

$$\text{Area of the 1}^{\text{st}} \text{ path} = 80 \times 4 = 320 \text{ cm}^2 \text{ and}$$

$$\text{Area of the 2}^{\text{nd}} \text{ path} = 60 \times 4 = 240 \text{ cm}^2$$

$$\text{Area common to both the paths} = 4 \times 4 = 16 \text{ cm}^2$$

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$$\therefore \text{Total area of both Paths} = 320 + 240 - 16 = 544 \text{ cm}^2$$

$$\therefore \text{Cost of gravelling} = 544 \times 12 = \text{Rs } 6,528$$

38. Ans: 70m, 220 m; Area of a circle = πr^2

$$\Rightarrow 3850 = \frac{22}{7} \times r \times r \quad \Rightarrow r = 35 \text{ m}^2$$

$$\therefore \text{Diameter} = 2 \times 35 = 70 \text{ m}$$

$$\text{Now, Circumference of a circle} = 2\pi r = 2 \times \frac{22}{7} \times 35 = 220 \text{ m}$$

39. Ans: 28m, 88 m ; Area of a circle = πr^2

$$\Rightarrow 616 = \frac{22}{7} \times r \times r \quad \Rightarrow r = 14 \text{ m}^2 \quad \therefore \text{Diameter} = 2 \times 14 = 28 \text{ m}$$

$$\text{Circumference of a circle} = 2\pi r = 2 \times \frac{22}{7} \times 14 = 88 \text{ m}$$

40. Ans: 1078 m²; Area of the lawn without path = $\pi r^2 = \frac{22}{7} \times 21 \times 21 = 1386 \text{ m}^2$ and

$$\text{Area of the lawn including path} = \pi r^2 = \frac{22}{7} \times 28 \times 28 = 2464 \text{ m}^2$$

$$\therefore \text{Area of the path} = 2464 \text{ m}^2 - 1386 \text{ m}^2 = 1078 \text{ m}^2$$

41. Ans: 2310 m²;

$$\text{Area of the lawn without path} = \pi r^2 = \frac{22}{7} \times 49 \times 49 = 7546 \text{ m}^2 \quad \text{Area of the lawn including path} = \pi r^2 = \frac{22}{7} \times 56 \times 56 = 9856 \text{ m}^2$$

$$\therefore \text{Area of the path} = 9856 \text{ m}^2 - 7546 \text{ m}^2 = 2310 \text{ m}^2$$

42. Ans: 176 m, 28 m; Circumference – radius = 148

$$\Rightarrow 2\pi r - r = 148 \quad \Rightarrow r(2\pi - 1) = 148$$

$$\Rightarrow r \left(2 \times \frac{22}{7} - 1 \right) = 148 \quad \Rightarrow r \left(\frac{44}{7} - 1 \right) = 148$$

$$\Rightarrow r \left(\frac{37}{7} \right) = 148 \quad \Rightarrow r = \frac{148 \times 7}{37} = 28 \text{ m}$$

$$\therefore \text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 28 = 176 \text{ m}$$

43. Ans: 132 m, 21 m; Circumference – radius = 111

$$\Rightarrow 2\pi r - r = 111 \quad \Rightarrow r(2\pi - 1) = 111$$

$$\Rightarrow r \left(2 \times \frac{22}{7} - 1 \right) = 111 \quad \Rightarrow r \left(\frac{44}{7} - 1 \right) = 111$$

$$\Rightarrow r \left(\frac{37}{7} \right) = 111 \quad \Rightarrow r = \frac{111 \times 7}{37} = 21 \text{ m}$$

$$\therefore \text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 21 = 132 \text{ m}$$

44. Ans: 270 cm²;

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Area of the parallelogram = $30 \times 9 = 270 \text{ cm}^2$

45. **Ans: 160 cm^2 ;**

Area of a parallelogram = $ah = 20 \times 8 = 160 \text{ cm}^2$

46. **Ans: $1,764 \text{ cm}^2$;** Area of a circle = $\pi r^2 \Rightarrow 1386 = \pi r^2$

$$\Rightarrow r^2 = 441 \quad \Rightarrow r = 21 \quad \Rightarrow \text{Diameter} = 21 \times 2 = 42$$

Since diameter of the circle = side of the square,

$$\text{area of the square} = a^2 = 42^2 = 1764 \text{ cm}^2$$

47. a) **Ans: $12,474 \text{ cm}^2$**

Largest circle which can be cut will have a diameter of 126 cm (*as 126cm is the smallest side of the rectangle*)

$$\therefore \text{Radius } \frac{126}{2} = 63 \text{ cm}$$

$$\therefore \text{Area of circle} = \pi r^2 = \frac{22}{7} \times 63 \times 63 = 12,474 \text{ cm}^2$$

b) **Ans: $19,026 \text{ cm}^2$;** Area of the board = $126 \times 250 = 31500$

$$\therefore \text{Area of the waste board} = 31500 - 12474 = 19026$$

48. **Ans: Rs. 16,000**

Length of the plot = 20 m and Area of the plot = 320 m

$$\therefore \text{Width of the plot} = \frac{320}{20} = 16 \text{ m}$$

Length of the plot without the pathway = 16 m

Width of the plot without the pathway = 12 m

$$\therefore \text{Area of the plot without the pathway} = 16 \times 12 = 192 \text{ m}^2$$

$$\therefore \text{Area of the path} = 320 - 192 = 128 \text{ m}^2$$

$$\therefore \text{Cost of constructing the pathway} = 128 \times 125 = \text{Rs. } 16,000$$

49. **Ans: 1 m;** Area of the lawn with path = 150 m^2

Let the width of the path = x

$$\therefore \text{Length of the lawn with path} = 15 + 2x$$

$$\text{And Width of the lawn with path} = 10 + 2x$$

$$\therefore \text{Area of the path} = (15 + 2x)(10 + 2x) - 150 = 54$$

$$\Rightarrow 150 + 30x + 20x + 4x^2 - 150 = 54$$

$$\Rightarrow 50x + 4x^2 = 54 \quad \Rightarrow 4x^2 + 50x + 54 = 0$$

$$\Rightarrow 2x^2 + 25x + 27 = 0 \quad \Rightarrow 2x^2 + 27x - 2x + 27 = 0$$

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$$\Rightarrow x(2x + 27) - 1(2x + 1) = 0$$

$$\Rightarrow (x - 1) = 0 \Rightarrow x = 1\text{m.}$$

50. **Ans: 150 m², Rs. 4,950;**

Area of the lawn with path = $23 \times 12 = 276 \text{ m}^2$

Area of the lawn without path = $18 \times 7 = 126 \text{ m}^2$

Area of the path = $276 - 126 = 150 \text{ m}^2$

Cost of constructing the path = $33 \times 150 = \text{Rs. } 4,950.$